Changes in Vertical Jump Height, Anthropometric Characteristics, and Biochemical Parameters After Contrast Training in Master Athletes and Physically Active Older People

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Abstract

González-Ravé, JM, Delgado, M, Vaquero, M, Juarez, D, and Newton, RU. Changes in vertical jump height, anthropometric characteristics, and biochemical parameters after contrast training in master athletes and physically active older people. *J Strength Cond Res* 25(7): 1866–1878, 2011—The purpose of this study was to determine the effects of 16 weeks of contrast training (CT) on older adults (with different levels of physical conditioning) in vertical jump performance (squat jump [SJ], countermovement jump [CMJ], CMJ during 15 seconds [CMJ15], depth jump [DJ]), body weight, fat percentage, muscle mass (MM), muscle cross-sectional area ([CSA] of the arm and thigh) and biochemical parameters (creatine kinase [CK], creatinine, and urea). Sixteen older (63.55 $\pm$ 6.89 years) recreational master runners (A) and 16 physically active older people (60.30 $\pm$ 5.18 years) though not athletes (NA), participated in the CT using a combination of heavy-resistance and explosive exercise. The dependent variables were measured pretraining and posttraining. The CT resulted in significant improvements ($\alpha = 0.05$) for both groups in jump performance. The SJ height improved in NA by 21.68% and in A by 21.81%, the CMJ height increased in NA by 21.51% and in A by 14.81%, the DJ height increased in NA by 26.45% and in A by 7.43%, and CMJ15 increased in NA by 6.20% and in A by 6.17%). Significant improvements in MM (16.44% in NA and 14.78% in A), thigh CSA (19.68% in NA and 21.67% in A), and arm CSA (7.43% in NA and 5.52% in A), and significant decreases in creatinine (8.65%) and CK (25.49%) in A were observed. In conclusion, CT improved vertical jump performance and MM in both groups, regardless of the training history and current physical activity of each group. These improvements were accompanied by a slight decrease in body fat but no changes in total body weight. These findings suggest that CT can have a significant effect on maximal jump height and MM in NA and A.

Key Words power, muscle mass, creatine kinase, resistance training, elderly

Introduction

It has been well documented that under normal conditions, the amount of muscle mass (MM) and absolute voluntary strength decrease after middle age, especially at the onset of the sixth decade in both sexes (9–11,29). Although MM decreases as a result of the aging process, this decline is considerably exacerbated by the sedentary lifestyle increasingly adopted by older adults (39). These age-related declines are not surprising, because aging is associated with lifestyle changes such as reductions in the quantity and intensity of physical activity, a decrease in MM, in metabolic and cardiovascular function and in neural function (22). However, Klitgaard et al. (22) found only strength trained elderly athletes to be stronger than untrained controls, with no difference between swimmers, runners, and untrained age peers.

Strength training in elderly men and women improves neuromuscular and morphological factors, which lead to increases in maximal force and explosive strength production (9,12,13,18,32,34). Current recommendations for resistance training in adults encourage slow-velocity contractions at

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25(7)/1866–1878

Journal of Strength and Conditioning Research

a relatively high percentage of maximal force (50–80%) to improve muscle strength (1). Importantly, increases in lower body strength are usually accompanied by significant improvements in the performance of activities of daily living (10). Further, muscle power has emerged as an important predictor of functioning and disability in older adults (34). It is therefore of great importance to investigate all aspects of exercise prescription for optimizing improvements in structure and function of older adults to maintain their quality of life.

Several studies have investigated heavy-resistance training or power training to develop strength in elderly people, but only a few studies have used contrast training (CT) (12,13,19) comparing middle-aged and older people. No studies have investigated the effects of CT on elderly subjects with different levels of physical fitness (recreational master runners and physically active older people though not athletes). Examination of these differences could be of great importance to identify possible changes related to the previous level of physical fitness. Based on this assumption, the purpose of this study was to identify the adaptations induced after CT on jump capacity, body composition, and physiological markers.

The term "contrast training" refers to a workout that involves the use of exercises of contrasting loads, that is, alternating heavy and light exercises, set for set (7). In this study, we developed a modified CT program from Häkkinnen et al. (12) based upon their recommended duration, repetitions (volume), load and velocity (intensity), and the frequency of performance (density). It is clear that CT has profound effects on muscle size, strength, and power (38) Furthermore, CT has proven to be safe and highly effective for increasing power and MM in elderly people (12,13,19).

Several studies have demonstrated functional improvements associated with changes in body composition. Improvements in functional independence in the elderly may substantially affect body composition, and help reduce the high incidence of obesity in this age group (4). Likewise, training programs administered to older adults may cause changes in body composition, which might contribute to functional improvement (9,32,35,36).

Although there has been much research on the issue, physiological markers have only been able to confirm an overtrained state (2). Creatine kinase (CK) has often been used as a marker of muscle cell membrane disruption after resistance training (26). Hurley et al. (18) and Clarkson and Temblay (5) showed CK levels were higher before and during the initial days of training than in the latter days and after training. In some instances, blood urea has been used in sport programs to help monitor training (2). Given the relatively intense nature of CT, these measures were taken to assess damage and recovery in these older participants and assess whether prior history of physical activity had any influence.

The hypothesis was that CT would alter vertical jump performance and anthropometry in endurance master athletes and physically active people. Additionally, we hypothesized that a history of competitive training and racing would attenuate the decline in strength that has been previously highlighted in noncompetitive aging adults.

The objective of this study was to examine changes in the following parameters in 60- to 70-year old athletes and nonathletes before and after CT: (a) squat jump (SJ), countermovement jump (CMJ), CMJ during 15 seconds (CMJ15), and jumping from the ground landing onto the platform, and then immediately performing a vertical jump (depth jump [DJ]); (b) body composition (weight, percent fat, MM, arm and thigh cross-sectional area [CSA]); (c) biochemical parameters (CK, creatinine, and urea).

Methods

Experimental Approach to the Problem

A pre–post-training intervention design was employed with an athletic and nonathletic group using CT for 16 weeks. Assessments were performed during 1 day and completed in an identical manner before and after training.

The present investigation was designed to study the effects of a 16-week CT program on vertical jump performance, anthropometry and blood markers of damage in 60- to 70-year-old athletes and nonathletes. Subjects were evaluated twice: at the beginning of the training (pretest) and after the 16-week training intervention (posttest). Our independent variable was the 16-week CT. Our dependent variables were vertical jump performance (SJ, CMJ, CMJ15, jump starting from the ground jumping onto the landing platform, and then performing a vertical jump for maximum height (DJ), body weight, fat percentage, MM, muscle CSA of the arm and thigh, and biochemical parameters (CK, creatinine, and urea).

Subjects

A population sample of 32 men divided into 2 groups, athletes, $n = 16$ (mean age $\pm$ *SD* 63.55 $\pm$ 6.89 years) and nonathletes, $n = 16$ (60.30 $\pm$ 5.18 years) were recruited. The master athletes group (group A) was composed of competitive, amateur athletes who had 15–17 years' experience with a year round endurance training schedule of 2 $h \cdot d^{-1}$ including warm-up and cool-down, 6 $d \cdot wk^{-1}$. During the treatment, group A maintained their scheduled training. All of the participants in group A were involved in continuous training methods for development of aerobic endurance (low intensity and medium volume). The physically active participant but not athlete group (group NA) was made up of volunteers taken from the physical activity program for elderly people run by a local City Council. The level of physical activity of these participants was low and intermittent, consisting of twice weekly, 30-minute sessions. All subjects were informed of the experimental risks and signed an informed consent document before the investigation. The investigation was approved by an ethics committee of the Granada University. None of the subjects had previously performed any resistance training (Table 1).

TABLE 1. Descriptive characteristics of group A and group NA.*

Characteristics	Recreational master athletes	Physically active people
Age (y)†	63.55 ± 6.89	60.30 ± 5.18
Physical activity level	Endurance master athletes	Community physical activity program
Training days	2 $h \cdot d^{-1}$, 6 $d \cdot wk^{-1}$	30 $min \cdot d^{-1}$, 2 $d \cdot wk^{-1}$
Training organization	Athletic training program	Sporadic practice
Competition	Yes	No

*Group A = athletic subjects to recreational master athletes; NA group = nonathletic subjects to physically active older people.
†Values are given as mean ± *SD.*

All subjects underwent a medical examination before participating in the study to assess their health and to detect any medical condition that could result in injury during the study. The criteria for selection of the participants were the following: men between 60 and 70 years of age with no blood pressure abnormalities and no evidence of respiratory disease. At the time of the study, subjects also needed to have normal joint mobility and no evidence of osteoporosis and arthritis. They also had to be free of vertigo related to brisk physical activity and rapid changes in position.

Procedures

Training Program. The strength training program used in this study was a combination of heavy and light resistance, known as "contrast training" (7). The contrast loading method involves the execution of alternating sets of an exercise with a heavy load followed by a set of the same exercise with a light load. Each training session included 6 exercises: chest press, military press, biceps curl using free weights, leg press and leg curl using machines, and lower abdominal exercises. The subjects trained 3 times per week on nonconsecutive days for 16 weeks. Each training session included a low-intensity warm--up, cycling for 4 minutes, followed by 3 minutes of static stretching, and approximately 50 minutes for the contrast training. Participants in all training sessions were under the close supervision of 2 qualified exercise leaders (graduates in Physical Activity and Sports Sciences) to ensure proper technique and to minimize the risk of injury. The first month of training included 3 sets of 8–10 repetitions 40–60% 1-repetition maximum (1RM) load and 2 sets of 4–6 repetitions 65% 1RM load. After the adjustment period, from weeks 5 to 12, training comprised 3 sets of 12–15 repetitions 45–50% 1RM load and 2 sets of 6–8 repetitions 70–75% 1RM load. During weeks 13–16, the training program consisted of 3 sets of 12–15 repetitions 45–50% 1RM load and 2 sets of 4–6 repetitions 80% 1RM load. The workloads (intensity) were adjusted monthly.

Repetition number and weight lifted were entered into the following Lander prediction equation (23): 1RM = 100 × rep

TABLE 2. Single-measure intraclass correlation coefficients of the dependent variables.*

Variable	Intraclass correlation	Confidence 95% Lower bound	Confidence 95% Upper bound	F test with true value 0 Value	df1	df2	Significance
SJ	0.77	0.46	0.87	6.47	31	32	0.00
CMJ	0.82	0.63	0.92	10.58	31	32	0.00
CMJ15	0.82	0.63	0.92	10.58	31	32	0.00
DJ	0.84	0.67	0.904	2.911	31	32	0.015
Weight	0.99	0.97	0.99	202.53	31	32	0.000
Muscle mass (Martin)	0.77	0.54	0.90	8.02	31	32	0.000
Muscle mass (Doupe)	0.66	0.34	0.84	4.85	31	32	0.000
CSA thigh	0.52	0.14	0.76	3.15	31	32	0.005
CSA arm	0.50	−0.18	0.69	2.28	31	32	0.03
Fat percentage	0.92	0.81	0.96	23.44	31	32	0.000
Creatine kinase	0.52	0.13	0.76	3.11	31	32	0.005
Creatinine	0.77	0.54	0.90	7.94	31	32	0.003
Urea	0.62	0.29	0.82	4.33	31	32	0.001

*CMJ = countermovement jump; DJ = depth jump; SJ = squat jump.

Table 3. Effect size of the dependent *t*-test.*†

Dependent variables	*r*	Sig.
SJ	0.94	0.000
CMJ	0.95	0.000
DJ	0.94	0.000
CMJ15	0.80	0.000
Fat percentage	0.933	0.000
Weight	0.99	0.000
Muscle mass (Martin)	0.94	0.000
Muscle mass (Doupe)	0.87	0.000
CSA thigh	0.75	0.000
CSA arm	0.72	0.000
Urea	0.63	0.002
Creatinine	0.88	0.000
Creatine kinase	0.65	0.001

*CMJ = countermovement jump; DJ = depth jump; SJ = squat jump; r = Pearsońs correlation coefficient.
†If $r > 0.5$, this represents a very large effect.

weight/(101.3 − 2.67203 × reps). The Lander equation has a high correlation in the following exercises: hip flexion, r: 0.746; biceps curl r: 0.869; hip abduction, r: 0.748; supine leg press, r: 0.795, or bench press, r: 0.901. During the 16-week experimental training period, subjects attended an average of 95% of the sessions as recorded in a training diary. All subjects also continued their physical activities such as walking, running, or gymnastics 1–2 times per week as they did before this study. The athletic group maintained their usual training schedule throughout the study.

Testing Procedures

Subjects performed a standardized warm-up at the start of the session, which consisted of 5 minutes of cycling (light aerobic) followed by 3 minutes of rest for all groups. The participants were instructed to "cycle at a comfortable speed," which was observed to be a consistent pace. All participants then performed a familiarization session during the morning, which consisted of 4 vertical jump tests. In the afternoon, 4 vertical jump types were performed with performance evaluated using an Ergojump Bosco System©, which recorded flight time (seconds), and jump height (cm). The subjects' hands were kept on their hips during all jump tests. Three maximal jumps were recorded, and the best maximum in terms of height was taken for further analysis. The rest period between jumps was always 30 seconds. Each subject was given external encouragement throughout all the tests.

Tests

The SJ began from a squatting start position with 90° knee angle. The subject was instructed to jump vertically for maximum height without any prior countermovement. If a countermovement was observed, the trial was not counted.

In the CMJ, the subject stood in an upright position, flexed the knees and hips into a squat position, and then immediately attempted to jump for maximum height.

Subjects were instructed to complete a series of vertical jumps from the force plate over a period of 15 seconds (CMJ15) landing and then immediately attempting to jump for maximum height.

In the final jump, subjects started from the ground jumping onto the landing platform and then performed a vertical jump for maximum height. This was termed the DJ. Subjects did not drop down from a bench as is usual for DJ because of concerns over injury.

Table 4. Results of the jump height tests before (pre) and after (post) strength training for group NA and group A.*†

		NA group	A group	*p* Value
SJ (cm)	Pre	12.69 ± 7.01	18.76 ± 3.20	<0.01‡
	Post	16.19 ± 7.69§	23.69 ± 5.36§	<0.05‖
CMJ (cm)	Pre	13.90 ± 7.95	21.28 ± 3.63	<0.01‡
	Post	17.71 ± 8.11§	24.78 ± 4.56§	<0.01‖
CMJ15 (%)	Pre	68.10 ± 6.59	75.60 ± 7.48	<0.001‡
	Post	74.22 ± 8.45§	81.77 ± 7.49§	<0.05‖
DJ (cm)	Pre	13.65 ± 8.33	22.04 ± 3.43	<0.01‡
	Post	18.56 ± 8.19¶	23.81 ± 4.07§	<0.05‖

*CMJ = countermovement jump; DJ = depth jump; SJ = squat jump; A group = athletic subjects to recreational master athletes; NA group = nonathletic subjects to physically active older people.
†Values are mean ± *SD*.
‡Significant differences between NA and A groups in pretest.
§Significant differences between pre and post: $p < 0.001$.
‖Significant differences between NA and A groups in posttest.
¶Significant differences between pre and post: $p < 0.01$.

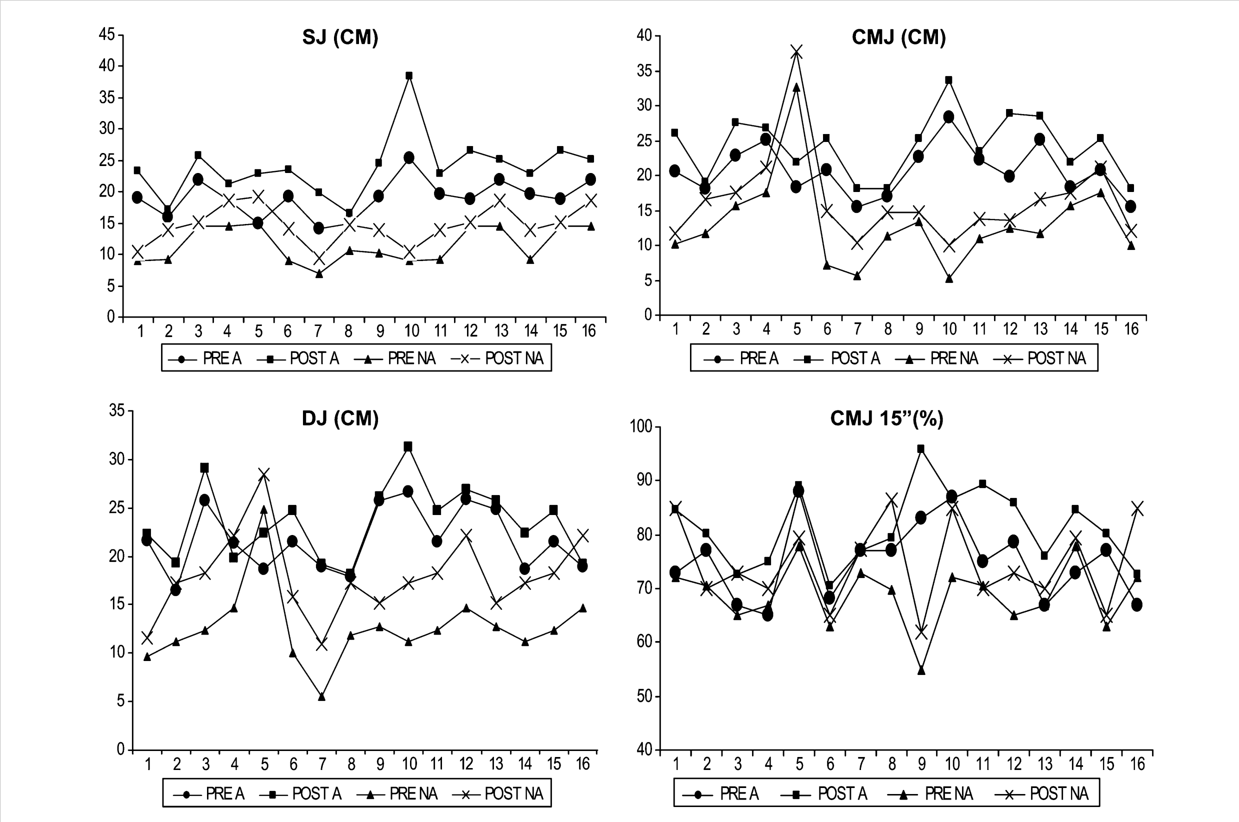

Figure 1. Spaghetti graph showing individual response of squat jump (SJ), countermovement jump (CMJ), depth jump (DJ), and CMJ during 15 seconds (CMJ15) in athletic subjects to recreational master athletes (A) and in nonathletic subjects to physically active older people (NA) before and after contrast training.

The vertical jump was chosen as a procedure for measuring the ballistic strength of the leg extensors (3,32). The use of the vertical jump to assess the response to training is widespread in the literature, including studies with the elderly (3,13,19). In addition, the use of vertical jumps provides further insight into the force capabilities of the leg extensor muscles (28). Moir et al. (28) suggest the assessment of vertical jump in athletes and recreationally active men can produce a high degree of reliability.

Anthropometry. On arrival at the laboratory, subjects were weighed to the nearest 0.1 kg using a Seca scale (Hamburg, Germany). The anthropometric assessments included body fat percentage (%BF) and arm and thigh CSA to assess differences in MM and to determine changes in fat % and body weight. In general, anthropometric measurements were performed using the following devices as established by the International Working Group of Kinanthropometry. A caliper (Holtain Ltd., Crymich, United Kingdom), with an accuracy of 0.1 mm, was used to determine %BF with skinfold measurements (abdomen, suprailiac, subscapular, triceps, midthigh, and calf). An anthropometric tape was used to measure perimeters and to determine the center between 2 anatomic landmarks. It is flexible, nonelastic, measures <7 mm in width, has a nongraduated space before the scale, and an easy to read centimeter scale (with an accuracy of 1 mm). A graduated compass, with an accuracy of 1.5 mm between its calipers was used to measure small diameters.

Muscle mass was measured by means of the equations of Martin et al. (25) and Doupe et al. (6). The CSA was measured by using Housh et al.'s equation (17) for the thigh and Heymsfield's (15) for the arm.

The %BF values were calculated using Peterson et al.'s equation (30). $\%BF = 20.94878 + (\text{age} \times 0.1166) - (\text{height} \times 0.11666) + (\text{sum4} \times 0.42696) - (\text{sum4}^2 \times 0.00159)$ where height is in centimeters and sum4 is the sum of the triceps, subscapular, suprailiac, and midthigh skinfold thicknesses.

The equation to estimate MM is $MM = STAT\ (0.0553CTG2 + 0.0987FG2 + 0.0331CCG2) - 2445$ ($SEE = 1.53$ kg, $r^2 = 0.97$), where STAT is stature (cm), CTG is thigh circumference corrected for the front thigh skinfold thickness (cm), FG is the uncorrected forearm circumference (cm), and CCG is the calf circumference corrected for the medial calf skinfold thickness (cm).

The equation of Doupe et al. (6) is as follows: $MM\ (g) = Ht\ (0.031MUThG^2 + 0.064CCG^2 + 0.089CAG^2) -$

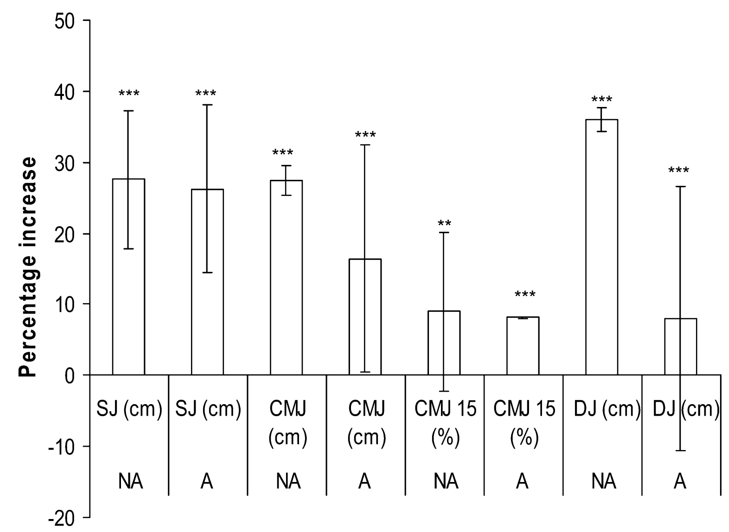

Figure 2. Mean percentage increase in each form of jump test. Significantly different: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

3.006. Variables (in cm) were Ht, height; MUThG, modified upper thigh girth; CCG, corrected calf girth; and CAG, corrected arm girth. The predictive ability of this equation was comparable to that of the original equation of Martin et al. (25) and was a valuable tool for MM estimation of male subjects in the 1981 Canada Fitness Survey. The upper-arm muscle plus bone area (17) was predicted using the equation $Am = (Ca - \Pi St)^2/4\Pi$, where midarm circumference (Ca) is corrected for the contribution of subcutaneous adipose tissue based on the midarm triceps skinfold thickness (St).

Finally, the thigh muscle area (17) was predicted using the equation $AMT = (4.68 \times PMM) - (2.09 \times Plma) - 80.99$, where PMM means midthigh circumference and Plma means midthigh skinfold thickness.

The circumferences were measured as follows: (a) *Thigh circumference*: Maximum thigh perimeter measured with the subject standing, legs slightly apart and weight evenly distributed. The anthropometrist kept the tape perpendicular to the long axis of the femur, measuring from the right side. (b) *Leg circumference*: Maximum circumference of the right leg measured with the subject standing, legs slightly part and weight evenly

Table 5. Results of the body composition parameters before (pre) and after (post) strength training for group NA and group A.*†

		NA group	A group	*p* Value
Height (cm)	Pre	162.66 ± 7.43	171 ± 5.76	
Weight (kg)	Pre	85.25 ± 19.04	77.91 ± 8.60	
	Post	85.66 ± 19.31	76.86 ± 8.22	
Fat % (anthropometric)	Pre	27.89 ± 9.77	16.10 ± 2.56	<0.001‡
	Post	26.09 ± 10.11	14.48 ± 2.63	<0.001§
Muscle mass (Martin) (kg)	Pre	30.47 ± 7.83	33.42 ± 4.73	
	Post	34.46 ± 8.27‖	37.12 ± 4.90‖	
Muscle mass (Doupe) (kg)	Pre	22.51 ± 6.86	27.79 ± 4.85	<0.05‡
	Post	26.94 ± 8.03‖	32.59 ± 4.62‖	<0.05§
CSA thigh (cm^2)	Pre	145.6 ± 17.93	148.8 ± 13.14	
	Post	157.3 ± 18.90¶	157.5 ± 11.02#	
CSA arm (cm^2)	Pre	31.46 ± 10.16	37.73 ± 5.77	
	Post	39.17 ± 11.77#	48.17 ± 9.90‖	

*A group = athletic subjects to recreational master athletes; NA group = nonathletic subjects to physically active older people; CSA = cross-sectional area.
†Values are mean ± *SD*.
‡Significant differences between NA and A group in pretest.
§Significant differences between NA and A group in posttest.
‖Significant differences between pre and post: $p < 0.001$.
¶Significant differences between pre and post: $p < 0.05$.
#Significant differences between pre and post: $p < 0.01$.

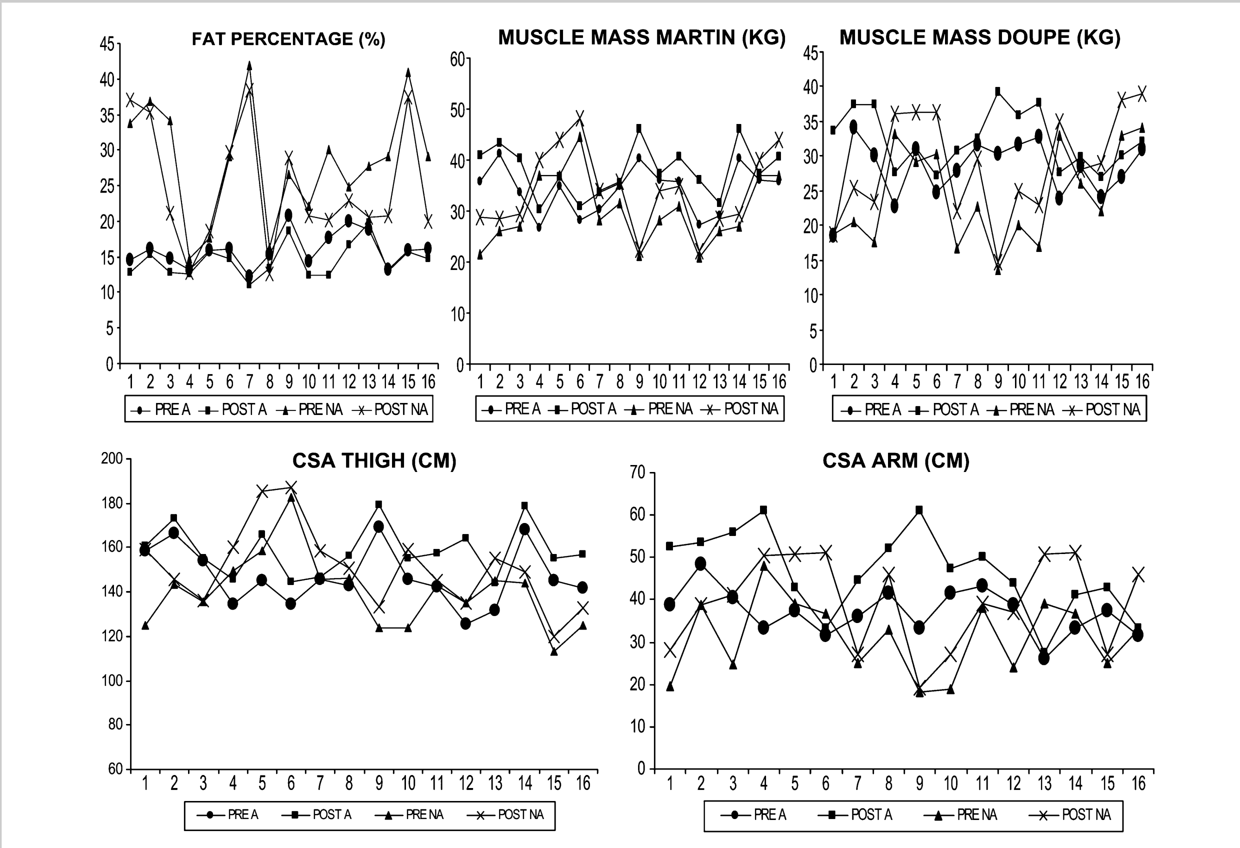

Figure 3. Spaghetti graph showing individual response of body composition in athletic subjects to recreational master athletes (A) and in nonathletic subjects to physically active older people (NA) before and after contrast training.

distributed. (c) *Forearm circumference*: the maximum circumference of the forearm. The subject extended the elbow with the forearm muscles relaxed and the arm in supination. The anthropometrist recorded the maximum circumference of the forearm, which is normally not more than 7 cm below the head of the radius.

All anthropometric dimensions were taken by the same tester who had previously demonstrated test–retest reliability of $r > 0.90$.

Blood Analysis

Urea, creatinine, and CK were measured from morning, fasting blood samples taken in a local hospital before and after the training program. Blood was drawn from an antecubital arm vein into 10-ml vacutainer tubes with no additives and allowed to clot at room temperature. Blood was centrifuged at 4°C and the serum stored at −70°C until analysis. Samples were analyzed using spectrophotometry (Hitachi 747 model, Tokyo, Japan).

Statistical Analyses

SPSS 20.0 for Windows was used to calculate mean ± *SD* and other statistical parameters. The test–retest reliability was examined using a single-measure intraclass correlation coefficient (ICC). The ICC was obtained for each batch of repeated measures, confirming the high reliability of the data gathering process. For example, for the SJ variable, 3 sets of 10 repeated measures were obtained in the first 4 weeks of the experiment. A value of 0.975 for the ICCr shows the reliability of the process. In the other sets of repeated measures, the ICCr was always >0.85. The Fleiss evaluation (8) defines concordance of variables as excellent when ICC is >0.75, good when it is 0.60–0.74, acceptable 0 when it is 0.40–0.59, and poor when it is <0.40. Thus, in this study, there were 8 variables with an excellent ICC (SJ, CMJ, CMJ15, DJ, MM (Martin), weight, fat percentage, and creatinine), 3 with an acceptable one (CSA thigh, CSA arm, and CK), and only 1 with a good one (MM [Doupe]). (Table 2).

The Shapiro–Wilks statistical test was used to determine the data normality. Student's *t*-test for independent samples was used to compare initial values between the athletic and nonathletic groups. The paired *t*-test gave the statistical significances of the differences between before and after training measurements. The effect size was examined using Pearson's correlation coefficient (Table 3). Probability limits

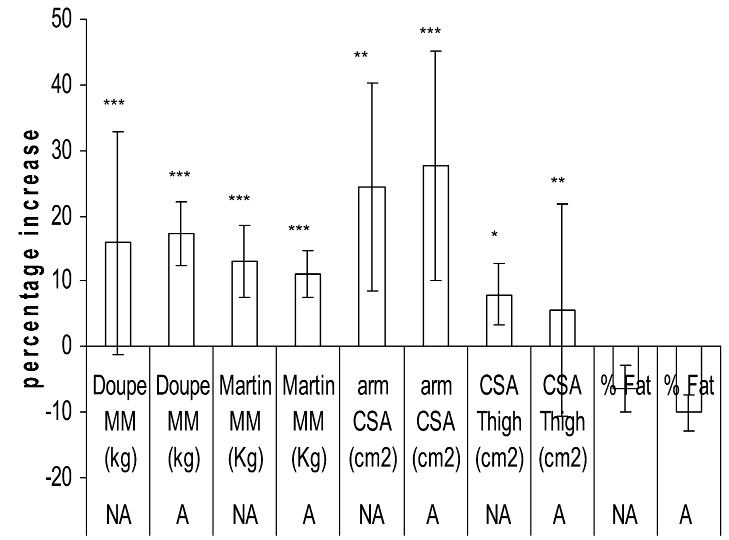

Figure 4. Mean percentage changes in body composition. Significantly different (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$).

were calculated (p values). Analysis of covariance (ANCOVA) was used to determine between-group differences, using the vertical jump height (CMJ, SJ, and DJ) as the covariates.

Results

Results of the jump height tests before and after the training intervention are presented in Table 4 and Figure 1 for both groups. The initial values for each test were significantly higher ($p < 0.05$) in group A than in group NA (Table 4).

After 16 weeks of training (Figure 2), the SJ, CMJ, DJ, and CMJ15 height increased for both groups.

The SJ height improved in NA by 21.68% ($p < 0.001$) and in A by 21.81% ($p < 0.001$). Large increases were also evident for CMJ after 16 weeks in NA of 21.51% ($p > 0.001$) and in A of 14.81% ($p < 0.001$). The DJ height increased in NA by 26.45% ($p < 0.001$) and in A by 7.43% ($p < 0.001$). Finally, the CMJ15 increased in NA by 6.20% ($p < 0.01$) and in A by 6.17% ($p < 0.001$). After the training program, the values for each of the jump height tests continued to be higher in group A than in group NA.

Pre and poststudy body composition information is given in Table 5 and Figure 3.

The initial values of fat percentage as measured by anthropometry were significantly lower in A by 42.27% ($p < 0.001$). The MM was significantly higher in group A by 18.99% ($p < 0.05$) vs. in group NA. The other parameters did not differ significantly; however, weight was higher in group NA by 8.60%. There were insignificant changes in body mass over the training period. The MM increased after 16 weeks of training in NA by 16.44% (Doupe's equation; $p < 0.001$) and by 11.57% (Martin's equation; $p < 0,001$.), and in A by 14.78% (Doupe's equation; $p < 0.001$) and 9.97% (Martin's equation; $p < 0.001$) (Figure 4).

Finally, both groups showed a significant increase in arm and thigh CSA after contrast training. Arm CSA increased during the 16 weeks of training in NA by 7.43% ($p < 0.01$) and in A by 5.52% ($p < 0.001$); and thigh CSA increased in NA by 19.68% ($p < 0.05$) and in A by 21.67% ($p < 0.01$) (Figure 4). After the study, fat percentage values continued to be significantly lower in group A and MM values (by the Doupe equation) continued to be significantly higher.

Results of the biochemical parameters before and after training are presented in Table 6 and Figure 5.

Table 6. Results of the biochemical parameters before (pre) and after (post) strength training for group NA and group A.*

		NA group	A group
Urea (mg·dL^{-1})	Pre	38.88 ± 4.10	39.84 ± 8.14
	Post	42.00 ± 6.40	36.46 ± 7.91
Creatinine (mg·dL^{-1})	Pre	1.02 ± 0.17	1.13 ± 0.14
	Post	1.00 ± 0.15	1.04 ± 0.10†
Creatine kinase (U·L^{-1})	Pre	167.22 ± 81.38	250.46 ± 108.7
	Post	122.88 ± 46.45	186.61 ± 91.38†

* Values are means ± *SD*.
A group = athletic subjects to recreational master athletes; NA group = nonathletic subjects to physically active older people.
†Significant differences between pre and post: $p < 0.05$.

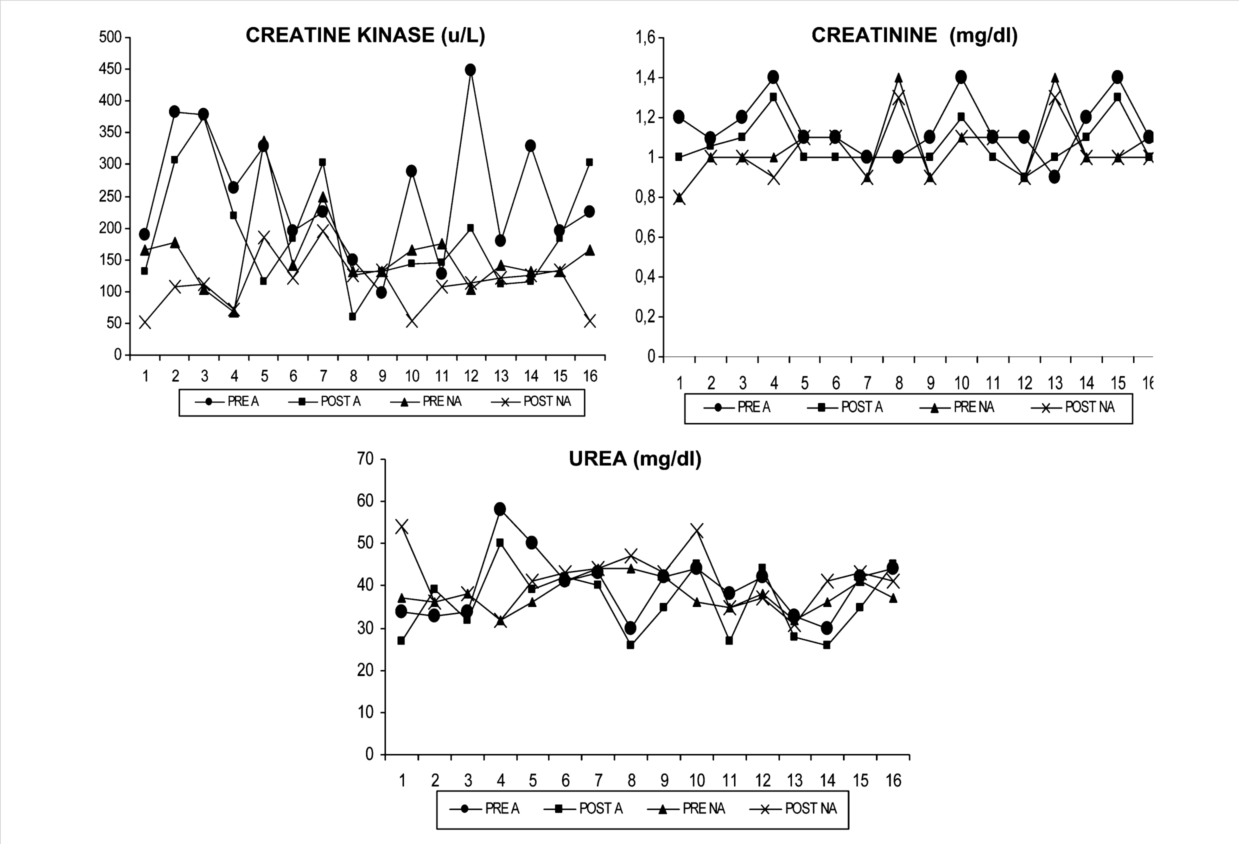

Figure 5. Spaghetti graph showing individual response of biochemical parameters in athletic subjects to recreational master athletes (A) and in nonathletic subjects to physically active older people (NA) before and after contrast training.

The initial values for each of the biochemical parameters did not differ between groups. However, training resulted in a significant decrease ($p < 0.05$) in creatinine and CK in group A (Figure 6).

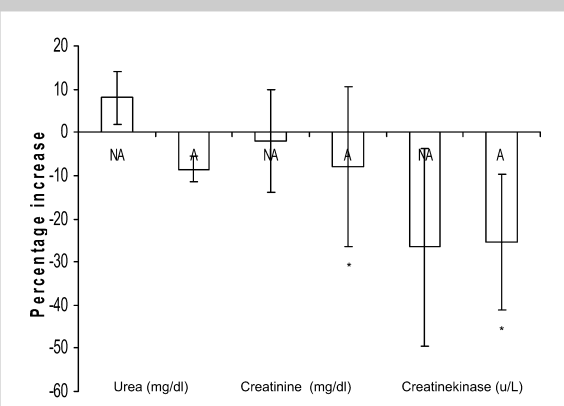

Figure 6. Mean percentage changes in biochemical parameters. Significantly different (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$).

After the training program, the values for each biochemical parameter did not differ between groups.

The repeated-measures ANCOVA showed a nonsignificant effect of vertical jump height (CMJ, SJ, and DJ) on the results obtained on MM and CSA with the training program performed.

Discussion

The purpose of this study was to investigate changes in jump height, anthropometry, and blood markers of damage in 60- to 70-year-old endurance athletes and nonathletes before and after a specific type of exercise purported to be effective for increasing strength and power called contrast training. We also assessed whether prior exercise history and current training status of older subjects would protect them against strength and power loss and maintain healthy body composition. Finally, given the well-established interference effects of aerobic endurance exercise on MM and strength gains, we aimed to determine if contrast training would still be evident in older endurance athletes. To the best of our knowledge, no other study has examined the effects of contrast training in master athletes and physically active older people.

The results obtained showed that strength training with the contrast method improved the force and power produced by the legs as measured by jump height. This change was similar to that in other methods habitually used in strength training (18). Strength training, using the contrast method, significantly improved ($p < 0.001$) explosive force in the lower extremities as measured by SJ, CMJ DJ, and CMJ15 for both groups. Our hypotheses were that contrast training would alter vertical jump performance in endurance master athletes and physically active people; these results support our hypotheses; no significant differences were found between groups in the improvements achieved in the jumping tests. The improvement obtained in SJ for both groups agrees with results achieved by Frontera et al. (9), Tracy et al. (38), and Häkkinen et al. (12). Some neurological changes occur as strength increases, but after 4–5 weeks of training, muscle hypertrophy prevails over neural factors (29). After the contrast training, we observed highly significant increases ($p < 0.001$) in CMJ for A (14.81%) and NA (21.51%), which is similar to the study by Kalapotharakos et al. (20). An increase in DJ values in this study implies gains in strength in the elderly for both groups, particularly for group NA (26.45%). It would appear that the neuromuscular system is able to tolerate the higher eccentric loads being generated and contribute to increasing the jump height for both groups. After the training program, both groups showed significant reductions in fatigue in the CMJ15 test, in agreement with Hicks and McCartney (16). It is unclear whether this was because of the number of repetitions per session and thus the overall increased workload or specifically because of the contrast training. Further comparisons are not possible because the number of studies using fatigue testing in elderly people is very small (only Meltzer [27] used an alactic capacity test).

Among the elderly, not only strength but also endurance athletes seem to have greater muscle strength than untrained men (36,37). Baseline test results for SJ, CMJ, and CMJ15 were significantly higher in group A, which implies that the level of previous endurance training influences explosive force in older people, above that obtained by simply being physically active. On the other hand, higher values in the CMJ15 in group A may be because of better neuromuscular performance properties including higher fatigue resistance and better intra and intermuscular coordination because of previous endurance training. Additionally, we hypothesized that a history of competitive training and racing would attenuate the decline in strength that has been previously highlighted in noncompetitive aging adults; these results support our hypotheses. However, group NA demonstrated significantly greater increases in CMJ and DJ performance compared to group A perhaps indicating the principle of diminishing returns where the A group had less scope for improvement because of their higher pretest performance.

It has been demonstrated that concurrent strength and endurance training appears to inhibit strength development when compared with strength training alone, although the nature of this inhibition is not clear (14,24). The study by Hennessy and Watson (14) in master endurance athletes showed that concurrent strength and endurance training does appear to elicit a different adaptive response than either strength or endurance performed in isolation. Although the combined training may interfere with muscle hypertrophy in aging men (21), in our study with older athletes, the contrast training increased explosive force of the lower extremities in the athletes who were completing a considerable volume of endurance exercise throughout the intervention period. There was no difference between the 2 groups, so interference did not appear to occur.

Body mass changes from ageing are because of the accumulation and distribution of body fat (9,32,37), loss of muscular mass, reduction of bone density, and decrease in basal metabolic energy expenditure. These changes can be modified by physical activity. Before starting the contrast training program, the groups showed insignificant weight differences, despite the fact that group A, which participated in aerobic activities, had a lower percentage of body fat. The insignificant differences in body weight between both groups were fundamentally because of the previous inactivity and sedentary lifestyle of subjects in group NA. These participants had a greater index of overweight, and the main interest of the study was to compare subjects with low levels of physical activity with trained subjects.

The higher MM in group A could be explained by their athletic activity (the main factor, and the amount of minutes of training, 20 $h \cdot wk^{-1}$ in A vs. 1 hour in NA), which provided a more intensive stimulus than that provided by walking (in group NA), which does not appear to prevent MM loss to the same degree (10). The absence of changes in body weight demonstrated by NA subjects and those in group A have been previously reported in similar studies (10,12,18,36). We hypothesized that a history of competitive training and racing would attenuate the decline in strength that has been previously highlighted in noncompetitive aging adults, these results support our hypotheses.

After a 16-week training period groups A and NA experienced significant decreases ($p < 0.05$) in fat tissue as measured by skinfolds, which partly coincides with the results obtained by Sipilä and Suominen (36) and Sipilä (35). This could be explained by the fact that contrast training probably results in considerable energy expenditure during the session and because of the high intensity, there is most likely an elevation of metabolic rate for 1–3 days after the exercise session. This feature has not been studied in our work, but Pollock et al. (31) have analyzed it in their longitudinal study and showed that strength training in master athletes maintained their MM but increased fat percentage 2–2.5% per year because body fat increases significantly with age. Pollock et al. (31) showed that the physiological capacities of older athletes are reduced despite continued vigorous endurance exercise over a 20-year period

(8–15% per decade). Changes in body composition appeared to be less than those shown for the healthy sedentary population and were related to changes in training habits. Body weight remained stable for the athletes, but they had to have trained regularly for at least the preceding 2 years, placed first, second, or third in regional, national, or international competitions in running and be athletes who performed frequent moderate-to-rigorous endurance training. Regardless of physical activity level, percent fat increased 2–2.5% per decade in both groups.

In our study, contrast training produced highly significant and meaningful increases in MM for both groups as have been reported by previous research into traditional strength training (12,36). Although some studies have reported that contrast training increases MM (9,12,18) in young people, other research noted no hypertrophy, possibly because of the short training time (33). It has been demonstrated that endurance training may blunt the hypertrophic response to strength training (24), but in our study with older athletes, concurrent strength and endurance training resulted in similar increases in arm and thigh CSA as in the NA group which only strength trained.

These results agree with those of previous studies (9,13,18) mainly in those including contrast training in the session (13,19). As such, appreciable improvements in MM, accompanied by an increase in CSA can explain the improvement in force production obtained in this study.

The results of the ANCOVA showed that ability of vertical jump in pretest had a nonsignificant effect on the MM changes with the training program performed.

Before the study, there were no significant differences between the groups in any of the biochemical parameters (urea, creatinine, and CK). However, CK tended to be slightly higher in group A ($p = 0.06$), which could have been because of either a larger training load or a lack of adaptation because it was the beginning of the sports season (9). Creatine kinase can be measured as either an acute response or an adaptation to training. In our study, CK was monitored all through the 16-week-training period. The CK values showed that the muscle Z lines were damaged, results which occur in young, adult and elderly people (18,26,33). Resistance training workouts evoke a decrease in CK, which agrees with the data of Hurley et al. (18) and Clarkson and Temblay (5) whose study showed that CK levels were higher before and during the initial days of training than in the latter days and after training. There may be an adaptive mechanism (owing to alterations of the cell membrane) or a reinforcement of muscle fiber with connective tissue, therefore making muscle fiber more resistant to damage. Zajaç et al. (40) indicated that Serum CK activity can be used as a marker of muscle damage, whereas changes in CK levels after exercise correlate significantly with the rate of regeneration of muscle cells. Clarkson and Temblay (5) observed a relationship between the increased activity of CK and muscle tissue damage, yet the underlying mechanism was unexplained. It is well known that serum CK activity is elevated for several days after intensive physical exercise and that training induces adaptive changes that allow for decreased CK levels under a standard load (40). Values $>200\ U \cdot L^{-1}$ can indicate an excessive workload, and as a consequence, it may be advisable to propose recuperation training with reduced loading.

From the current results, it can be concluded that recovery capacity between training loads is faster for both groups at the end of the intervention, as shown by CK levels (30). Creatinine levels decreased significantly in group A and not significantly in the NA group, reflecting adaptive responses to training. Creatinine blood levels in both groups may have been the result of muscle adaptations to a training program, which led to a decrease of this blood metabolite. On the other hand, a decrease in creatinine levels below normal may imply muscle atrophy, which is not reflected in the present study because of the increase in MM and CSA. After the training program, creatinine levels were more or less the same in both groups, and urea was not significantly altered ($p > 0.1$), but was higher in group NA. Given that there were no differences before the study, this would indicate a slight fatigue level or a slower adaptation to training in NA.

In conclusion, strength training using contrast methods increased vertical jump and MM in endurance athletes and physically active subjects. These improvements took place accompanied by a slight decrease in body fat, without changes in total body weight. These effects are produced by an adaptive response and without acute fatigue or overtraining, as shown by the biochemical parameters.

Practical Applications

The main finding was that master athletes and physically active older people, at least within the present training period of 16 weeks, have a similar ability to increase muscle power in a functional activity like the vertical jump, and MM in response to a periodized contrast training program. Any practical application requires careful implementation and individual experimentation in older people. Correctly designed and competently supervised, this contrast training program carries no extra overload on skeletal muscle development in older men as proved by an adaptive response and without acute fatigue or overtraining, as shown by the biochemical parameters. This research is important because of the demonstrated benefits of increased strength and power for function and quality of life of older people. Moreover, master runners should have strength conditioning in their training programs to avoid damage of tendons and muscle wasting that it is produced with only running endurance training. The present results support the premise that loss of strength and MM may vary in relation to the level of physical conditioning (recreational master runners and physically active older people though not athletes). These findings suggest that for activities involving muscle function related to SJ, CMJ, and DJ; contrast training can have a significant effect on maximal jump height and MM in physically active older subjects and master athletes. This training method provides an interesting variation that appears well tolerated and effective for this age group regardless of previous training status.

Acknowledgments

We would like to thank all the athletes and physically active subjects who participated as subjects in this study. This study was partially supported by a grant from the Cordoba City Council, Spain.

References

1. American College of Sports Medicine. *ACSM's Guidelines for Exercise Testing and Prescription* (6th ed.). Baltimore, MD: Wolters Kluwer, 2000.
2. Bacharach, DW, Petit, M, and Rundell, KW. Relationship of blood urea nitrogen to training intensity of elite female biathlon skiers. *J Strength Cond Res* 10: 105–108, 1996.
3. Bosco, C, Luhtanen, P, and Komi, PV. A simple method for measurement of mechanical power in jumping. *Eur J Appl Physiol* 50: 273–282, 1983.
4. Cavani, V, Mier, CM, Musto, AA, and Tummers, N. Effects of a 6-week resistance-training program on functional fitness of older adults. *J Aging Phys Activ* 10: 443–452, 2002.
5. Clarkson, PM and Temblay, I. Exercise induced muscle damage, repair and adaptation in humans. *J Appl Physiol* 65: 1–6, 1998.
6. Doupe, MB, Martín, AD, Kriellars, DJ, and Giesbrecht, GG. A new formula for population-based estimation of whole body muscle mass in males. *Appl Physiol Nutr Metab* 22: 598–608, 1997.
7. Duthie, GM, Young, WB, and Aitken, DA. The acute effects of heavy loads on jump squat performance: An evaluation of the complex and contrast methods of power development. *J Strength Cond Res* 16: 530–538, 2002.
8. Fleiss, JL. The design and analysis of clinical experiments. New York, NY: Wiley, 1986.
9. Frontera, W, Meredith, CN, ÓReilly, KP, Knuttgen, HG, and Evans, WJ. Strength conditioning in older men: Skeletal muscle hypertrophy and improved function. *J Appl Physiol* 64: 1038–1044, 1988.
10. Galvão, DA and Taaffe, DR. Resistance training for the older adult: Manipulating training variables to enhance muscle strength. *Strength Cond J* 27: 48–54, 2005.
11. Häkkinen, K, Alen, M, Kallinen, M, Izquierdo, M, Jokelainen, K, and Lassila, H. Muscle CSA, force production, and activation of leg extensors during isometric and dynamic actions in middle-aged and elderly men and women. *J Aging Phys Activ* 6: 232–247, 1998.
12. Häkkinen, K, Kallinen, M, Izquierdo, M, Jokelainen, K, Lassila, H, and Malkia, E. Changes in agonist–antagonist EMG, muscle CSA, and force during strength training in middle-aged and older people. *J Appl Physiol* 84: 1341–1349, 1998.
13. Häkkinen, K, Kraemer, W, Newton, R, and Alen, M. Changes in electromyographic activity, muscle fibre and force production characteristics during heavy resistance/power strength training in middle-aged in older men and women. *Acta Physiol Scand* 171: 51–62, 2001.
14. Hennessy, LC and Watson, AWS. The interference effects of training for strength and endurance simultaneously. *J Strength Cond Res* 8: 12–19, 1994.
15. Heymsfield, SB, McManus, C, Smith, J, Stevens, V, and Nixon, DW. Anthropometric measurement of muscle mass: Revised equations for calculating bone-free arm muscle area. *Am J Clin Nutr* 36: 680–690, 1982.
16. Hicks, AL and McCartney, N. Gender differences in isometric contractile properties and fatigability in elderly human muscle. *Can J Appl Physiol* 21: 441–454, 1996.
17. Housh, DJ, Housh, TJ, Weir, JP, Weir, LL, Johnson, GO, and Stout, JR. Anthropometric estimation of thigh muscle cross-sectional area. *Med Sci Sport Exerc* 27: 784–791, 1995.
18. Hurley, BF, Redmond, RA, Partley, RE, Treuth, MS, Rogers, MA, and Goldberg, AP. Effects of strength training on muscle hypertrophy and muscle cell disruption in older men. *Int J Sports Med* 16: 378–384, 1995.
19. Izquierdo, M, Häkkinen, K, Ibañez, J, Garrues, J, Anton, A, and Zuñiga, A. Effects of strength training on muscle power and serum hormones in middle–aged and older men. *J Appl Physiol* 90: 1497–1507, 2001.
20. Kalapotharakos VI, P, Tokmakidis, S, Smilios, I, Ichalopoulos, M, Liatis, J, and Godolias, G. Resistance training in older women: Effect on vertical jump and functional performance. *J Sports Med Phys Fitness* 45: 570–575.
21. Karavirta, L, Häkkinen, A, Sillanpää, E, García-López, D, Kauhanen, A, Haapasaari, A, Alen, M, Pakarinen, A, Kraemer, WJ, Izquierdo, M, Gorostiaga, E, and Häkkinen, K. Effects of combined endurance and strength training on muscle strength, power and hypertrophy in 40-67-year-old men. *Scand J Med Sci Sports* 2011. [Epub ahead of print].
22. Klitgaard, H, Mantoni, M, Schiaffino, S, Ausoni, S, Gorza, L, Laurent-Winter, C, Schnohr, P, and Saltin, B. Function, morphology and protein expression of ageing skeletal muscle: A cross-sectional study of elderly men with different training backgrounds. *Acta Physiol Scand* 140: 41–54, 1990.
23. Lander, J. Maximum based on repetitions. *Nat Strength Cond Assoc J* 6: 60–61, 1985.
24. Leveritt, M, Abernethy, P, Barry, B, and Logan, P. Concurrent strength and endurance training. *Sports Med* 28: 413–427, 1999.
25. Martin, AD, Spenst, LF, Drinkwater, DT, and Clarys, JP. Anthropometric estimation of muscle mass in men. *Med Sci Sport Exerc* 22: 729–733, 1990.
26. McBride, JM, Radzwich, R, Mangiuçno, L, McCormick, M, McCormick, J, Volek, J, Bush, JA, Augilera, B, Newton, RU, and Kraemer, WJ. Responses of serum creatine kinase activity to heavy resistance exercise in endurance and recreationally trained women. *J Strength Cond Res* 9: 139–142, 1995.
27. Meltzer, DE. Body-mass dependence of age related deterioration in human muscular function. *J Appl Physiol* 80: 1149–1155, 1996.
28. Moir, G, Button, C, Glaister, M, and Stone, MH. Influence of familiarization of the reliability of vertical jump and acceleration sprinting performance in physically active men. *J Strength Cond Res* 18: 276–280, 2004.
29. Narici, MV, Roi, GS, Landoni, L, Minetti, AE, and Cerretelli, P. Changes in force, cross-sectional area and neural activation during strength training and detraining of the human quadriceps. *J Appl Physiol* 59: 310–319, 1989.
30. Peterson, M, Czerwinski, SA, and Siervogel, RM. Development and validation of skinfold-thickness prediction equations with a 4-compartment model. *Am J Clin Nutr* 77: 1186–1191, 2003.
31. Pollock, ML, Mengelkoch, LJ, Graves, JE, Lowenthal, DT, Limacher, MC, Foster, C, and Wilmore, JE. Twenty year follow-up of aerobic power and body composition of older track athletes. *J Appl Physiol* 82: 1508–1516, 1997.
32. Porter, MM. The effects of strength training on sarcopenia. *Can J Appl Physiol* 26: 123–141, 2001.
33. Roth, SM, Martel, GF, Ivey, FM, Lemmer, JT, Metter, J, and Hurley, BF. High-volume, heavy-resistance strength training and muscle damage in young and older women. *J Appl Physiol* 88: 1112–1118, 2000.
34. Sayers, SP. High-speed power training: A novel approach to resistance training in older men and women. A brief review and pilot study. *J Strength Cond Res* 21: 518–526, 2007.

35. Sipilä, S. Physical training and skeletal muscle in elderly women. A study of muscle mass, composition, fiber characteristics and isometric strength. Studies in sport, physical education and health. University of Jyväskyla, Jyväskyla, Finland, 1996.

36. Sipila, S and Suominen, H. Effects of strength and endurance training on thigh and leg muscle mass and composition in elderly women. *J Appl Physiol* 78: 334–340, 1995.

37. Sipilä, S, Viitasalo, J, Era, P, and Suominen, H. Muscle strength in athletes aged 70–81 years and a sample population. *Eur J Appl Physiol* 63: 399–403, 1991.

38. Tracy, BL, Ivey, FM, Hurlbut, D, Martel, GF, Lemmer, JT, Siegel, EL, Metter, EJ, Fozard, JL, Fleg, JL, and Hurley, BF. Muscle quality II. Effects of strength training in 65- to 75-yr-old men and women. *J Appl Physiol* 86: 195–201, 1999.

39. Vandervoort, AA and Symons, TB. Functional and metabolic consequences of sarcopenia. *Can J Appl Physiol* 26: 90–101, 2001.

40. Zajac, A, Waskiewicz, Z, and Pilis, W. Anaerobic power, creatine kinase activity, lactate concentration, and acid-base equilibrium changes following bouts of exhaustive strength exercises. *J Strength Cond Res* 15: 357–361, 2001.